Akoizo Original Practice - 2027 Rotation Form 5

Division C | Astronomy | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) Astronomical metals are…

Unit: Galaxies, stellar populations, and the interstellar medium.
Worked-case data: A region is dark in visible light but bright in infrared and contains embedded young sources. Explain without declaring it empty.

A. All solids only

B. Only charged atoms

C. Elements heavier than helium

D. Only iron and nickel

Answer: _____

2. (2 points) A giant can cool at the surface while brightening because…

Unit: Cloud collapse and mass-dependent stellar evolution.
Worked-case data: A toy main-sequence model uses L □ M^3·^5 and lifetime □ M/L. Compare a 2 Msun star with the Sun.

A. Temperature has no effect

B. Its area becomes zero

C. Its radius increases strongly

D. Distance must decrease

Answer: _____

3. (2 points) False-color blue necessarily means…

Unit: Observational astronomy and the electromagnetic spectrum.
Worked-case data: A star has parallax 50 mas and approximate blackbody temperature 5800 K.

A. A known temperature

B. A nearby object

C. Only the assigned display channel

D. Visible blue starlight

Answer: _____

4. (2 points) A flux-limited distant sample favors…

Unit: Advanced graph and image interpretation.
Worked-case data: Two stars have equal effective temperature but luminosities 1 and 100 Lsun. Compare radii.

A. Intrinsically luminous detectable objects

B. Every star equally

C. Only the lowest-mass stars

D. No selection effect

Answer: _____

5. (4 points) Calculate Rs for 10 Msun.

Unit: Supernovae, white dwarfs, neutron stars, and black holes.
Worked-case data: Use 3 km per solar mass as a rounded Schwarzschild-radius scale for a 10 Msun nonrotating black hole.

Answer: _____
_____
_____

6. (4 points) Why can a white dwarf be hot but faint?

Unit: Stellar properties, fusion, and the H-R diagram.
Worked-case data: A star has radius 2 Rsun and temperature 2 Tsun. Calculate L/Lsun.

Answer: _____
_____
_____

7. (4 points) Why not add magnitudes to combine stars?

Unit: Astrophysics equations and quantitative synthesis.
Worked-case data: A source has m = 10 and M = 5, with negligible extinction. Find distance and compare its flux with the same source at 10 pc.

Answer: _____
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8. (4 points) Why not model a galaxy as one star?

Unit: Galaxies, stellar populations, and the interstellar medium.
Worked-case data: A region is dark in visible light but bright in infrared and contains embedded young sources. Explain without declaring it empty.

Answer: _____
_____
_____

9. (4 points) Name two turnoff assumptions or corrections.

Unit: Cloud collapse and mass-dependent stellar evolution.
Worked-case data: A toy main-sequence model uses L □ M^3·^5 and lifetime □ M/L. Compare a 2 Msun star with the Sun.

Answer: _____
_____
_____

10. (6 points) Solve this case and explain your reasoning: A star has parallax 50 mas and approximate blackbody temperature 5800 K.

Answer: _____
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11. (2 points) A synthetic star has parallax 0.002 arcseconds. Use d(pc)=1/p to find distance.

Answer: _____
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12. (2 points) A standard-candle exercise gives apparent magnitude 15, absolute magnitude 5 and negligible extinction. Use m-M=5log10(d/10 pc). Find d.

Answer: _____
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13. (2 points) A star has twice the reference radius and the same temperature. Find its luminosity ratio using L proportional to R^2 T^4.

Answer: _____
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14. (2 points) Use Wien's constant 2.90e-3 m K for a 5,800 K star. Find peak wavelength.

Answer: _____
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_____

15. (2 points) An artificial galaxy recession speed is 350 km/s. Use H0=70 km/s/Mpc and the nearby linear Hubble law to find distance.

Answer: _____
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_____

16. (2 points) A normalized binary exercise has separation 2 AU and period 2 years. Use total mass in solar masses=a^3/P^2. Find total mass.

Answer: _____
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17. (2 points) NASA identifies M82 X-2 as an ultraluminous pulsar. What does a coherent rotation-linked pulse train imply about the compact object, and why is luminosity alone insufficient?

Answer: _____
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18. (2 points) SN 2014J is a Type Ia supernova in M82. Explain why standardized Type Ia luminosities can constrain distance and why dust must be considered.

Answer: _____
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19. (2 points) GW170817 was observed in gravitational waves and electromagnetic radiation. Explain the value of this multi-messenger evidence for a neutron-star merger.

Answer: _____
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20. (2 points) Synthetic starburst-galaxy case 5: ultraviolet emission and infrared dust emission both increase. Explain why using only visible light can miss part of the recent star formation.

Answer: _____
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